1. Incident Management Scenario:

An organization experiences a distributed denial of service (DDoS) attack, causing significant service outages. The incident response team has mitigated the attack, but several internal applications are still non-functional due to network congestion. As an incident manager, what is your next step?

A) Close the incident since the DDoS attack is mitigated.
B) Escalate the incident to problem management.
C) Prioritize restoring the internal applications.
D) Conduct a root cause analysis.

Answer: C
Explanation: After mitigating the attack, restoring functionality should be the next priority. Closing the incident prematurely (Option A) without restoring full functionality is incorrect. While problem management (Option B) and root cause analysis (Option D) are necessary, they come after ensuring business-critical services are restored.

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2. Problem Management Scenario:

A critical database server has experienced frequent crashes over the past month. Although incidents have been resolved each time, the issue continues to reoccur. As part of problem management, what should be the focus to prevent further disruptions?

A) Escalate to the database vendor for technical support.
B) Initiate a formal root cause analysis and create a problem ticket.
C) Continue to resolve incidents as they occur.
D) Increase server capacity to handle more traffic.

Answer: B
Explanation: Initiating a root cause analysis (RCA) through problem management is crucial in identifying and permanently resolving the underlying issue. Escalating to the vendor (Option A) may help but should be part of the RCA process. Continuously resolving incidents (Option C) without addressing the root cause will lead to recurring problems. Increasing server capacity (Option D) might not address the actual issue if it's not related to traffic.

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3. Capacity Management Scenario:

Your company's web application has seen a significant increase in traffic, leading to slower response times. Based on capacity management, which of the following should be done first to avoid further performance degradation?

A) Monitor the traffic to ensure it is legitimate and not a DDoS attack.
B) Add more servers to the web application cluster immediately.
C) Conduct a performance test to assess the impact of additional traffic.
D) Proactively scale the infrastructure based on projected traffic growth.

Answer: D
Explanation: Capacity management involves planning for future needs. Proactively scaling the infrastructure (Option D) is crucial to maintain performance during periods of growth. Monitoring for DDoS (Option A) is a valid concern but unrelated to long-term capacity issues. Adding servers

immediately (Option B) might be reactive and unplanned. Conducting performance tests (Option C) should be part of capacity planning but does not address immediate needs.

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4. Change Management Scenario:

A new security patch needs to be deployed across all critical servers. As part of change management, the patch deployment must go through a formal change process. What is the most important reason for this?

A) To inform all stakeholders about the patch deployment.
B) To ensure that the patch does not introduce new vulnerabilities.
C) To comply with organizational change control policies.
D) To avoid unplanned downtime during the patch deployment.

Answer: C
Explanation: Complying with the organization's change control policy (Option C) ensures that the patch is thoroughly tested and approved, avoiding unexpected issues. Informing stakeholders (Option A) and preventing new vulnerabilities (Option B) are important, but the formal change management process ensures that these are systematically addressed. Avoiding unplanned downtime (Option D) is one of the goals of change management.

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5. Incident Management Response:

A ransomware attack encrypts all the organization's files, and the attacker is demanding a ransom for decryption. What is the most appropriate incident management action?

A) Pay the ransom to restore operations quickly.
B) Disconnect affected systems from the network and engage law enforcement.
C) Attempt to restore from the latest backups.
D) Isolate the affected systems and begin negotiations with the attacker.

Answer: B
Explanation: Disconnecting the systems from the network and involving law enforcement (Option B) are crucial steps to contain the spread of the ransomware and follow legal protocols. Paying the ransom (Option A) is not recommended, as it may not guarantee file recovery. While restoring from backups (Option C) is ideal, it should only be done after ensuring the malware is fully contained.

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6. Problem Management:

After a network outage, the issue was traced back to a misconfiguration in a core router. Problem management is tasked with ensuring this does not happen again. What is the best approach?

A) Create a temporary workaround to restore service.
B) Implement a permanent fix and update the configuration management database (CMDB).
C) Involve the network team to monitor the router closely.
D) Document the incident and close the ticket.

Answer: B
Explanation: Implementing a permanent fix and updating the CMDB (Option B) ensures that the misconfiguration will not occur again. A temporary workaround (Option A) is only a stopgap, and monitoring (Option C) doesn't address the root cause. Closing the ticket (Option D) without fully resolving the issue is premature.

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7. Capacity Management Issue:

The IT department receives complaints about slow application performance during peak hours. What should capacity management do to handle future demand?

A) Migrate the application to a cloud provider that offers better performance.
B) Implement load balancing to distribute the traffic evenly.
C) Monitor the application's performance and take action when thresholds are breached.
D) Conduct a capacity forecast and implement changes to meet future demand.

Answer: D
Explanation: Conducting a capacity forecast (Option D) ensures that the organization is prepared to handle future demand. Migration (Option A) or load balancing (Option B) might be solutions, but they should come after the forecasting. Monitoring (Option C) without planning for future growth is reactive, not proactive.

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8. Change Management Conflict:

A major software release is scheduled for next week, but the security team has requested additional testing due to concerns about potential vulnerabilities. What should the change manager do?

A) Approve the release and proceed as scheduled.
B) Escalate the issue to the change advisory board (CAB).
C) Delay the release until all security concerns are addressed.
D) Implement the release and patch vulnerabilities later.

Answer: B
Explanation: Escalating to the CAB (Option B) allows for a balanced decision based on risk. Proceeding with the release (Option A) without resolving security concerns could expose the organization. Delaying the release (Option C) might be a valid outcome, but the CAB should make this decision. Patching later (Option D) is risky.

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9. Incident Response:

During an incident, a firewall begins blocking all traffic due to an incorrect rule implementation. What should the incident response team do first?

A) Remove the firewall rule to restore traffic flow.
B) Disable the firewall temporarily.
C) Roll back to the last known good configuration.
D) Reboot the firewall and monitor for issues.

Answer: C
Explanation: Rolling back to the last known good configuration (Option C) is the safest and most effective way to restore normal operations. Removing the rule (Option A) without understanding its full impact or disabling the firewall (Option B) creates further security risks.

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10. Problem Management Long-term Resolution:

A recurring issue with slow database queries has been traced back to poor indexing. What is the best approach to resolve this permanently under problem management?

A) Rewrite the queries to avoid the indexing problem.
B) Perform a thorough review and optimize the database indexing strategy.
C) Add more memory to the database server.
D) Implement a temporary solution and revisit the issue later.

Answer: B
Explanation: Optimizing the database indexing strategy (Option B) addresses the root cause of the problem and provides a long-term solution. Rewriting queries (Option A) might be a workaround but not an optimal solution. Adding memory (Option C) may improve performance but won't solve the indexing issue. A temporary solution (Option D) isn't ideal for long-term problem management.